

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	5

Case Study Topic: Snapdeal Product Catalog Sorting using Timsort

Word Done Summary:

In this case study, we analysed Snapdeal's product catalog sorting system using the Timsort algorithm. We studied how Timsort efficiently handles partially sorted datasets by identifying natural runs, reversing descending runs, and performing stable merges. We also examined insertion sort optimization, galloping mode, and compared Timsort with quicksort and merge sort for real-world workloads.

Implementation Details:

1. Represented catalog data using integer arrays.
2. Identified natural ascending and descending runs.
3. Reversed descending runs into ascending order.
4. Applied stable merging of sorted runs.
5. Used insertion sort for small segments.
6. Compared Timsort with merge sort and quicksort.
7. Studied galloping mode optimization.

Result:

Timsort efficiently sorted partially ordered product catalog data by leveraging natural runs and stable merging. The algorithm demonstrated superior practical performance on real-world partially sorted datasets.

Output:

```
Original Array:
3 7 11 5 4 6 9 12 1 2 8 10 13 14

Sorted Array using Timsort:
1 2 3 4 5 6 7 8 9 10 11 12 13 14
```

GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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